

Pawan Kumar

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Professional Summary

Backend-focused Software Engineer with experience building high-performance APIs, scalable AWS infrastructure, and AI-powered agentic systems in production environments. Skilled in Golang, cloud-native architectures, and LLM-based workflows including RAG, LangChain, and LangGraph to automate debugging, data access, and decision-making.

Education

Indian Institute of Technology (BHU), Varanasi
B.Tech in Electronics Engineering: GPA: 8.70 / 10

Nov 2020 – May 2024

A.S. Patna Central School, Patna
Intermediate (Class XII): 92%
Matriculation (Class X): CGPA: 10 / 10

Apr 2016 – Apr 2019

Experience

Samsung (SRID)

Software Engineer

Delhi, India

Jul 2024 – Present

- **Backend Engineering:** Accelerated API performance by **98%** (500ms to 10ms) via query optimization and connection pooling. Productionized 8+ REST APIs serving **50K+ daily requests**, and architected a centralized authentication and request validation service securing 6+ microservices.
- **AI & Agentic Systems:** Conceived and launched **ServiceHub**, an AI-driven platform leveraging **RAG** to resolve queries across 10GB+ of internal data. Engineered an autonomous **CloudWatch log-analysis agent** that reduced manual debugging by **80%**, alongside a secure Natural Language-to-SQL workflow with human approval gates.
- **AWS & Infrastructure:** Directed an AWS account migration via **Terraform**, reducing migration time by 2-3 weeks. Orchestrated zero-downtime PostgreSQL blue-green upgrades, optimized ECS, Lambda, and RDS workloads, and streamlined Jenkins CI/CD workflows to cut deployment cycles by **80%**.
- **Tech Stack:** Golang, Python, Java, Spring Boot, LangChain, LangGraph, Vector DB, SQL, AWS, Terraform, Jenkins

Samsung(SRID)

Software Engineer Intern

Delhi, India

May 2023 – Jul 2023

- **Recommendation Engine:** Engineered a movie recommendation system leveraging graph-based algorithms using **PyTorch Geometric** and **NetworkX** to improve content discoverability.
- **Computer Vision:** Developed a robust thumbnail similarity detection pipeline utilizing **OpenCV** and **Scikit-learn** to identify duplicate media assets.

Projects

Intelligent Data & Jewellery Analytics Platform

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- Developed a comprehensive platform integrating real-time analytics dashboards with automated billing and inventory tracking, successfully reducing manual operations by **70%+**.

AI-Driven SQL Generator & Executor

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- Built a multi-step reasoning workflow combining AI-driven SQL generation with an approval-based execution pipeline.
- Improved internal query productivity by **75%** by enabling secure, automated data retrieval across 10+ database tables.

Technologies

- **Languages & Frameworks:** Go, Python, Java, C++ , TypeScript, SQL, Spring Boot, React, FastAPI
- **AI & Agents:** LLMs, RAG, LangChain, LangGraph, Prompt Engineering, Vector DBs
- **Cloud & DevOps:** AWS (EC2, ECS, Lambda, RDS, S3, IAM, API Gateway), Terraform, Docker, Jenkins, CI/CD
- **Tools:** Git, GitHub, JMeter, Postman, Swagger, Jira

Achievements, Leadership & Continuous Learning

- **Awards & Recognition:** Received the **Best Performance Award** (Aug 2025) and **Best Growth Project Award** (2025) at Samsung for contributions to scalable infrastructure and AI tooling.
- **Leadership: Captain, Volleyball Team** — Represented the institute at the INTER IIT Sports Fest 2024.
- **Courses & Interests:** Agentic Systems, Generative AI, Prompt Engineering, AWS Skill Builder Labs, DSA.